

Preston Caldwell

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Education

Rensselaer Polytechnic Institute, Troy, NY

Bachelor of Science in Electrical Engineering, Computer & Systems Engineering

Focus in Robotics and Automation

Honors: Magna Cum Laude

May 2023

GPA: 3.82/4.00

Shaker High School, Latham, NY

Advanced Regents Diploma, Honor Graduate, Seal of Biliteracy in Spanish

June 2019

Relevant Coursework

Robotics I, II

Fall 2022, Spring 2023

- Theoretical and practical applications of Forward and Inverse Kinematics, Path Planning, PID Systems, and Object Manipulation. Programming in MATLAB and Python with YAHBOOM robots in a laboratory setting.
- Leveraged ROS OpenCV packages to create an autonomous convoy of robots communicating through light signals.

Mechatronics

Spring 2023

- Modeling and simulation of Dynamic Electro-Mechanical systems using MATLAB and Simulink.
- Created a self-balancing segway using State Feedback Control.

Experience

Software Engineer at *Amentum*

November 2023 - Present

- Created, delivered and maintained physics-based fluid models for Kennedy Space Center's Exploration Ground Systems group to train Firing Room operators under NASA's Artemis Program.
- Collaborated with internal customers and analyzed data to ensure models matched real system performance.
- Reduced team's time spent on repetitive, tedious work through several automated test and tool scripts.
- Assisted in upgrading team's electrical library for more robust and physically accurate models.

Software Engineer at *Lockheed Martin*

Summer 2021, Summer 2022, May - November 2023

- Experience with federated simulation systems in C++. Assisted in network infrastructure upgrades, wrote automated tests, developed software to interface with hardware components for the user.
- Acted as Lead Software Engineer for a team working on updates to an F-35 simulator maintenance device.
- Experience with basic computer networking, flight simulator software and hardware, and Unreal Engine 4.
- Gained exposure to the "Engineering V" development cycle, and AGILE mindset.

Projects

Home Server

November 2021 - Present

- Created a Debian-based home server using Docker Compose for personal cloud storage, gaming and streaming services. Used reverse proxy to direct traffic between different containers.

Skills

Engineering: Control Systems, Circuitry, Simulation & Analysis, FPGA Programming, Autonomous Systems

Software: Linux, Docker, NX, LTSPICE, Jenkins, Git, Java, Python, C++, C, MATLAB, Simulink, Excel

Clubs & Organizations

RPI Club Rugby

October 2019 – May 2022

